

Variational Equilibrium in Cognitive Systems: A Conceptual Framework for Adaptive Viability

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Abstract

Cognitive systems must acquire information, act effectively, and preserve the resources and internal structures on which future performance depends. These demands are addressed by optimization, active inference, resource-rational analysis, and physiological regulation, but their integration remains a useful design problem. This conceptual paper proposes the Unified Variational Intelligence Framework (UVIF) as an organizing account of five interacting influences: epistemic pull, selective protection, computational drag, action efficiency, and sustainability. We formulate these influences as compatible vector fields subject to explicit admissibility constraints and distinguish stationary equilibria from invariant operating sets and tracking under environmental change. A restricted analytical example shows both how a stable equilibrium can arise and how the proposed formulation can coincide with constrained optimization. The intended contribution is therefore an

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explicit, inspectable decomposition of regulatory demands, rather than a demonstrated alternative to optimization or a universal law of intelligence. We relate the framework to cognitive theories, present hypothetical applications, and specify comparative tests involving resource manipulation, deployment horizons, irreversible outcomes, and component ablation. No empirical validation is reported; companion simulations illustrate a stipulated policy model. The framework’s value depends on whether measurable, independently specified components improve prediction or system design relative to comparably resourced alternatives. Open questions concern identifiability, safety enforcement, stability under changing conditions, and the empirical justification of the five-component decomposition.

Keywords: Variational equilibrium, Cognitive systems, Bounded rationality, Active inference, Adaptive regulation, Viability

1. Introduction

Cognitive systems operating over extended periods must do more than perform an isolated task. They must decide what to learn, protect useful knowledge, allocate limited computation, act before opportunities disappear, and preserve resources for future operation. These demands interact: additional inference may improve a decision while delaying it, and protecting an established representation may preserve competence while impeding adaptation. The problem is therefore not simply to identify desirable outcomes, but to explain how a system regulates these competing demands over time.

Existing research already addresses important parts of this problem. Active inference relates perception and action through generative models and

epistemic and pragmatic value [1, 2]. Resource-rational analysis examines the allocation of limited computational resources [3, 4]. Homeostasis and allostasis describe regulation of viable internal conditions and anticipation
 15 of future demands [5, 6]. Safe reinforcement learning explicitly incorporates risk and constraints [7]. These approaches show that uncertainty, resource limits, and viability are established concerns, not considerations absent from optimization-based accounts.

The Unified Variational Intelligence Framework (UVIF) proposes an or-
 20 ganizing decomposition that brings five regulatory influences into a common description: epistemic pull, selective protection, computational drag, action efficiency, and sustainability. Its motivating question is whether keeping these influences explicit improves the explanation, diagnosis, or design of persistent cognitive systems. We use *force* for a contribution to state dynam-
 25 ics, not necessarily a physical force. We use *Variational Equilibrium* for an admissible stationary configuration of specified dynamics and distinguish it from the broader maintenance of viable operation.

Hypothesis 1.1 (Adaptive viability). *For some cognitive systems, explicitly modelling the interaction of information acquisition, protection, compu-
 30 tational cost, effective action, and future resource availability may explain adaptation under changing conditions more effectively than models that omit those interactions.*

This hypothesis neither states that all intelligence requires exactly five components nor rules out an optimization-based implementation. Persistence
 35 alone is also insufficient to establish intelligence: a passive stable system may persist without learning or flexible action. The proposed framework concerns

regulation within systems that already exhibit cognitive capabilities.

1.1. Contribution and scope

The paper makes three contributions. First, it defines a five-component
40 organizational vocabulary and gives a dimensionally compatible dynamical
formulation with explicit constraints. Second, it separates the proposed con-
tribution from related cognitive and control frameworks, including cases in
which the formulations coincide. Third, it develops hypotheses and an evalu-
ation protocol that could test whether the decomposition adds predictive or
45 engineering value.

This is a conceptual synthesis, not a systematic review or an empirical
validation study. The literature is selected to motivate the components and
examine neighbouring approaches; no exhaustive search or evidence-ranking
procedure is claimed. The analytical example establishes a property of one
50 stipulated model only. No experiments or simulation results are reported,
and cross-domain examples are analogies and design scenarios rather than
independent confirmations.

Section 2 clarifies the relation between optimization and viability. Sec-
tion 3 develops the natural and cognitive motivation. Section 4 specifies the
55 framework. Sections 5–7 examine domain translations, cognitive theories,
and hypothetical applications. Sections 8–11 address comparison, implica-
tions, limitations, and evaluation. Section 12 concludes.

2. Optimization, Regulation, and Viability

2.1. What the critique of optimization does and does not imply

60 Optimization provides a language for choosing actions or parameters under objectives and constraints. It does not inherently require a stationary environment, unlimited computation, complete information, or a single fixed goal. A scalar objective can encode multiple considerations, and multi-objective methods need not select one fixed scalarization. Likewise, sequential decision models can include resource stocks, irreversible events, and
65 deployment horizons in their state and transition descriptions.

The relevant difficulty is consequently one of model specification and implementation. An objective that omits an important consequence can incentivize undesirable behaviour; a constraint that is incorrectly estimated
70 can fail to protect the system. Safe reinforcement learning addresses such concerns through risk-sensitive objectives, constraints, and restrictions on exploration [7]. Alignment additionally raises questions about whose values and interests determine acceptable outcomes [8]. Renaming these considerations as forces does not resolve either problem.

75 UVIF instead emphasizes explicit representation of regulatory demands. Its potential benefit is that designers can inspect which mechanisms acquire information, preserve internal structure, consume computation, produce useful action, and maintain future capacity. Whether this decomposition is useful must be assessed against alternatives that receive the same information,
80 resources, and opportunities for adaptation.

2.2. *A motivating example: autonomous navigation*

Consider a mobile agent operating with limited battery capacity in an incompletely mapped environment. Additional sensing can reduce uncertainty but consumes time and energy. A short route can increase progress while reducing the reserve needed to recover from an obstacle or return to a charging station. Updating the map may improve navigation while introducing errors into previously reliable representations.

These trade-offs can be addressed through constrained planning or control. The UVIF description adds a proposed assignment of responsibilities: episodic pull evaluates useful sensing; selective protection regulates changes to critical representations; computational drag accounts for inference expenditure; action efficiency evaluates progress and timeliness; sustainability tracks future resource availability. Collision avoidance remains a separately enforced admissibility requirement, not a preference that a sufficiently large information benefit can outweigh.

The example illustrates the distinction between a soft influence and a hard constraint. Reducing a risk penalty is a preference; remaining inside a verified safe operating set is a dynamical property. A controller may have to sacrifice progress or stop when no safe action exists. Neither the existence of five named components nor a small net force certifies that property.

2.3. *Bounded information and timely action*

Partial observability and limited computation motivate a balance between further inference and action. They do not require an appeal to quantum uncertainty or a general claim that all uncertainty is irreducible. Some uncertainty is reducible by measurement; some persists because relevant vari-

ables are hidden, observations are noisy, or future disturbances are unknown. Resource-rational models already ask when further computation is worth its cost [3, 9].

In UVIF, bounded information motivates measurable stopping and allocation rules. A useful implementation should specify when additional information changes the action enough to justify its acquisition cost and delay. An assertion that the system has achieved “balance” is inadequate unless the underlying quantities and decision rule are identified.

3. Natural and Conceptual Foundations

3.1. Regulation in biological systems

Homeostasis concerns the regulation of internal conditions that support physiological function [5]. Adaptive homeostasis describes transient changes in regulatory capacity in response to signals [10], while homeodynamics emphasizes changing patterns of organization rather than residence at a single fixed state [11]. Allostasis adds anticipation: resources can be mobilized in response to predicted demand rather than only after a deviation occurs [6].

These accounts motivate a distinction between immediate performance and preservation of the capacity to perform later. They do not establish that biological regulation instantiates the five UVIF components as separable mechanisms. Biological systems can also fail, become trapped in maladaptive states, or persist under conditions that are undesirable for the organism. Natural persistence is therefore a source of hypotheses, not a guarantee of optimality, resilience, or ethical acceptability.

3.2. *Ecological resilience and the limits of analogy*

130 Ecological research describes resilience through feedback, reorganization, and responses to disturbance [12, 13]. This motivates studying operating regions and transitions rather than only average performance. However, the size of a basin of attraction is not a universal measure of resilience: conclusions depend on which variables, disturbances, timescales, and functions are
135 considered. A persistent ecological regime may also be undesirable from a conservation or human-welfare perspective.

Mapping ecological processes to epistemic or pragmatic terms requires particular care. An ecosystem need not possess a single policy, a unified goal, or a model of its environment. Such translations can organize a research ques-
140 tion, but they do not by themselves demonstrate cognition at the ecosystem level or a common mechanism across scales.

3.3. *What is variational about the proposal?*

In mechanics, a stationary-action formulation and local equations of motion can provide related descriptions under appropriate assumptions. Sta-
145 tionarity of an action does not imply that the physical state is stationary or asymptotically stable. Variational methods can also be used to derive optimization algorithms [14]; they do not oppose optimization as a class.

The name UVIF retains the motivating idea of interacting variations in regulatory demands. The general vector-field formulation below is not de-
150 rived from a common action functional and need not be a gradient flow. A strict variational interpretation applies only after a functional and the relevant geometry have been specified. Section 4.5 supplies a restricted potential-

based example and makes its equivalence to constrained optimization explicit. Establishing a more general variational derivation remains an open problem.

155 3.4. *Variability, adaptation, and normative choices*

Variability can support exploration and adaptation, but its consequences depend on the environment and the mechanisms that retain useful changes. Genetic variation provides one biological example [15]; it is not evidence that stochasticity always improves cognitive regulation. Deterministic controllers
160 may also adapt to changing inputs. The present framework does not rely on a contrast between deterministic failure and stochastic resilience.

Similarly, “nature-aligned” describes biological inspiration rather than a normative criterion. Human designers and affected stakeholders must specify which functions deserve protection, which resource costs matter, and what
165 outcomes are unacceptable. The framework makes these choices visible but cannot determine them from persistence alone.

4. The UVIF Organizational Framework

4.1. *State, fields, and admissible dynamics*

Let $\mathbf{s} \in \mathcal{S} \subseteq \mathbb{R}^n$ describe an explicitly chosen operating state. Depend-
170 ing on the application, its coordinates may include belief summaries, policy parameters, protected memory variables, and resource reserves. The state must contain the variables needed to model their evolution; otherwise the description may require additional memory or stochastic inputs. The following deterministic formulation is a modelling template, not a complete agent
175 implementation.

Write the unconstrained update field as

$$F(\mathbf{s}, t) = w_E \mathcal{E}_{\text{pull}} - w_D \mathcal{D}_{\text{drag}} - w_P \mathcal{P}_{\text{prot}} + w_A \mathcal{A}_{\text{eff}} + w_S \mathcal{S}_{\text{sus}}, \quad (1)$$

where each field is evaluated at $(\mathbf{s}, t)$ and each weight $w_i = w_i(\mathbf{s}, t) \geq 0$ is specified by the model. All weighted terms must be vectors in the same state coordinates and have units of state change per unit time. Raw information
 180 gain, latency, and energy consumption cannot be added without explicit scaling and a mapping into these coordinates. The signs are conventions: $\mathcal{D}_{\text{drag}}$ and $\mathcal{P}_{\text{prot}}$ are defined in the direction of increasing cost or damage, so their negative contributions oppose those changes.

State-dependent weights may encode resource depletion, urgency, or
 185 threat. Their update rules are part of the model and must be reported or estimated; dependence on state does not make them free of modelling assumptions. A fixed-weight model is a legitimate special case. A component may also vanish in a particular context, and balance does not require equal force magnitudes.

190 For a nonempty, closed, convex, time-independent admissibility set $\mathcal{C}_{\text{adm}}$, one candidate constrained dynamics is

$$\dot{\mathbf{s}} = G(\mathbf{s}, t) := \Pi_{T_{\mathcal{C}_{\text{adm}}}(\mathbf{s})} F(\mathbf{s}, t), \quad \mathbf{s}(0) \in \mathcal{C}_{\text{adm}}, \quad (2)$$

where $T_{\mathcal{C}_{\text{adm}}}(\mathbf{s})$ is the tangent cone of feasible directions and Π is Euclidean projection onto that cone. This is a first-order idealization in which the up-
 195 date velocity can be selected directly. Existence, uniqueness, and invariance require appropriate regularity of the field and a suitable solution concept for the projected dynamics. It does not automatically implement a feasi-

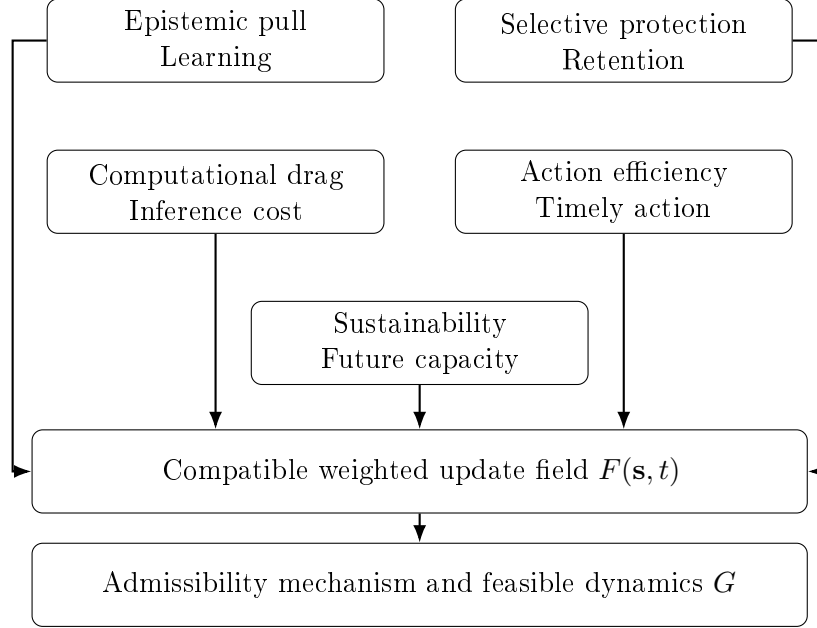


Figure 1: Five-component modelling architecture. Hard admissibility is distinct from soft protection and cost terms. Arrows indicate model composition, not a proven causal decomposition. Schematic prepared as editable TikZ code with OpenAI Codex (GPT-6) assistance.

ble controller for a plant with actuator limits, delays, inertia, or uncertain dynamics.

4.2. The five regulatory components

200 **Epistemic pull.** This component favours informative changes in the operating state. In a differentiable surrogate model one may set

$$\mathcal{E}_{\text{pull}}(\mathbf{s}, t) = \nabla_{\mathbf{s}} I(\mathbf{s}, t), \quad (3)$$

where I is expected information gain under a stated observation model and sampling policy. Expected information gain must identify the uncertainty being reduced; it is not interchangeable with confidence or task reward. For

205 discrete decisions, a gradient requires a justified relaxation or an alternative update rule. Curiosity and active inference provide motivation for this component [16, 17].

Selective protection. A candidate field is $\mathcal{P}_{\text{prot}} = \nabla_{\mathbf{s}} L_P$, where L_P measures damage to specified internal structures or loss of retained competence. This can describe regularization or resistance to updating protected parameters. It may reduce interference with previous learning, but it does not guarantee freedom from catastrophic forgetting. Protection also differs from a hard safety requirement: a soft penalty can be outweighed, whereas an admissibility constraint is intended to exclude a state or action.

215 **Computational drag.** A compatible cost-based field is

$$\mathcal{D}_{\text{drag}}(\mathbf{s}, t) = \nabla_{\mathbf{s}} C_{\text{comp}}(\mathbf{s}, t), \quad (4)$$

where C_{comp} is a differentiable surrogate for computation, latency, or memory expenditure. This is a cost gradient, not literal mechanical dissipation. A velocity-dependent damping term would require a different, explicitly specified dynamical model. Resource-rational analysis supplies an established basis for making computational expenditure part of decision-making [3].

Action efficiency. This component rewards useful and timely action. For a differentiable policy $\pi_{\mathbf{s}}$ and transition model p , define

$$J_A(\mathbf{s}, t) = \mathbb{E}_{a \sim \pi_{\mathbf{s}}, \mathbf{s}' \sim p(\cdot | \mathbf{s}, a)} [r(\mathbf{s}, a, \mathbf{s}') - c_{\text{act}}(\mathbf{s}, a)], \quad (5)$$

$$\mathcal{A}_{\text{eff}}(\mathbf{s}, t) = \nabla_{\mathbf{s}} J_A(\mathbf{s}, t). \quad (6)$$

Here r is an independently specified task benefit and c_{act} represents action cost or delay. Defining value only as proximity to an as-yet unknown equilibrium would be circular. This example is a local surrogate; a sequential

implementation must specify its planning horizon and account for state transitions. Costs assigned to this component should not also be counted in computational drag without an explicit reason.

Sustainability. This component represents prospective loss of operating
230 capacity, for example through

$$\mathcal{S}_{\text{sus}}(\mathbf{s}, t) = -\nabla_{\mathbf{s}} L_S(\mathbf{s}, t), \quad (7)$$

where L_S evaluates predicted depletion or degradation over a stated horizon. A penalty on instantaneous resource use alone does not establish long-term sustainability. A reserve b , for example, may need a balance law

$$\dot{b} = r_{\text{in}}(\mathbf{s}, a, t) - r_{\text{use}}(\mathbf{s}, a, t), \quad b \geq b_{\text{min}}, \quad (8)$$

with replenishment and consumption rates in compatible units. If b is part of
235 $\mathbf{s}$, this balance law must be implemented in the corresponding coordinate of the full dynamics; it cannot be an unrelated auxiliary equation. Projection alone cannot create an unavailable replenishment action.

The distinction between drag and sustainability is temporal and operational: drag concerns present inference expenditure, whereas sustainability
240 concerns future availability of required resources. Selective protection concerns damage to designated structures. These quantities can overlap, so the decomposition requires an accounting convention, normalization, and evidence that the components can be separately measured or manipulated.

4.3. Admissibility and safety enforcement

245 **Definition 4.1** (Admissibility set). $\mathcal{C}_{\text{adm}}$ is the set of states satisfying the physical and operational requirements chosen for a specified deployment con-

text. Where smooth inequality constraints suffice, write $\mathcal{C}_{\text{adm}} = \{\mathbf{s} : h_j(\mathbf{s}) \geq 0, j = 1, \dots, m\}$.

At a smooth boundary, an inward or tangent velocity satisfies $\nabla h_j(\mathbf{s})^\top \dot{\mathbf{s}} \geq 0$ for active constraints. Such conditions motivate projection and barrier-based safety mechanisms, but their applicability depends on the actual system dynamics [18]. A time-dependent constraint additionally involves $\partial h_j / \partial t$. Nonconvex domains, discrete actions, and uncertain disturbances require separate treatment.

Ethical and legal requirements can inform admissibility specifications, but they are not generally reducible to a complete set of state inequalities. A mathematical set certifies only the encoded model properties under its assumptions. It does not establish ethical acceptability or compliance by itself. Risk is handled through protection, explicit risk constraints, or the task model; it is not introduced as an unspecified sixth force.

4.4. Equilibrium, stability, and viable operation

Definition 4.2 (Stationary Variational Equilibrium). *For an autonomous constrained dynamics $\dot{\mathbf{s}} = G(\mathbf{s})$, the equilibrium set is*

$$\mathcal{VE}\Pi = \{\mathbf{s}^* \in \mathcal{C}_{\text{adm}} : G(\mathbf{s}^*) = 0\}. \quad (9)$$

A stationary Variational Equilibrium is any element of this set. Stability is an additional property, not a consequence of this definition.

An equilibrium is Lyapunov stable if, for every $\varepsilon > 0$, a sufficiently small initial displacement keeps the trajectory within ε for all future times. It is

locally asymptotically stable if it is stable and trajectories from a neighbourhood also converge to it. Initial states and neighbourhoods are understood
270 relative to $\mathcal{C}_{\text{adm}}$. For a nonisolated equilibrium set, convergence to the set need not imply convergence to a particular equilibrium.

At an interior equilibrium, $G = F = 0$. At a constrained boundary equilibrium, $G = 0$ can hold even when $F \neq 0$, because the unconstrained update points out of the feasible region. The equilibrium set need not be
275 smooth, connected, nonempty, or a manifold. Moreover, a trajectory cannot move along a set of stationary points under the same autonomous field while G remains zero everywhere on that trajectory.

An invariant operating set $K \subseteq \mathcal{C}_{\text{adm}}$ is a different object: trajectories starting in K remain there, potentially with $G \neq 0$. It can contain cycles or
280 other sustained activity. Viability asks whether admissible operation can be maintained by available controls, whereas invariance is relative to specified dynamics or a controller. Neither property is equivalent to zero net force or Pareto optimality.

For time-varying environments, the instantaneous zero set $\mathcal{V}\mathcal{E}\mathcal{H}(t)$ need
285 not itself be invariant. Tracking a selected $\mathbf{s}^*(t)$ requires a controller and bounds on disturbances and the rate of change of that reference. Accordingly, the broader UVIF design objective is adaptive viability, and stationary equilibrium is one restricted way to study it.

4.5. A restricted analytical example

290 Consider a dimensionless scalar operating variable $x \in [0, 1]$ in a fixed environment. Let $e, p, a, s \in [0, 1]$ be stipulated reference levels and let $\alpha_E, \alpha_P, \alpha_D, \alpha_A, \alpha_S > 0$ be rate coefficients. Define the weighted component

fields by

$$\begin{aligned}
w_E \mathcal{E}_{\text{pull}} &= \alpha_E(e - x), & w_P \mathcal{P}_{\text{prot}} &= \alpha_P(x - p), \\
w_D \mathcal{D}_{\text{drag}} &= \alpha_D x, & w_A \mathcal{A}_{\text{eff}} &= \alpha_A(a - x), \\
w_S \mathcal{S}_{\text{sus}} &= \alpha_S(s - x).
\end{aligned} \tag{10}$$

These linear fields are illustrative choices, not measured descriptions of cognition. Writing $k = \alpha_E + \alpha_P + \alpha_D + \alpha_A + \alpha_S$, their net field is $F(x) = k(x^* - x)$, where

$$x^* = \frac{\alpha_E e + \alpha_P p + \alpha_A a + \alpha_S s}{k}. \tag{11}$$

The stated assumptions imply $x^* \in [0, 1]$. The field points inward or is tangent at both endpoints, so no projection correction is needed. For any $x(0) \in [0, 1]$,

$$x(t) = x^* + [x(0) - x^*]e^{-kt}, \tag{12}$$

which remains in the interval and converges exponentially to x^* . Equivalently, $V = (x - x^*)^2/2$ satisfies $\dot{V} = -k(x - x^*)^2$. This proves stability only for this stipulated scalar example.

The same dynamics is gradient descent on

$$\begin{aligned}
\Phi(x) &= \frac{1}{2}[\alpha_E(x - e)^2 + \alpha_P(x - p)^2 + \alpha_D x^2 \\
&\quad + \alpha_A(x - a)^2 + \alpha_S(x - s)^2].
\end{aligned} \tag{13}$$

Thus its equilibrium also minimizes a scalar objective on $[0, 1]$. The example demonstrates that a force decomposition can be compatible with optimization; it cannot establish superiority over it. More generally, fixed weights and compatible gradient fields can yield a common potential, whereas state-dependent weights need not. Determining the relevant case is a property of a specified implementation.

Table 1: Illustrative domain translations; none constitutes validation of UVIF.

Domain	Candidate operating variables	Main translation problem
Physiology	Internal variables, regulatory signals, metabolic reserves	Distinguishing established feedback mechanisms from the proposed five components
Ecology	Population densities, resource stocks, interaction strengths	Avoiding an assumed unified agent, policy, or goal
Human cognition	Belief uncertainty, response time, retained knowledge	Separating latent components from correlated behavioural measurements
Artificial agents	Policy state, memory, latency, energy reserves	Accounting for controller cost and enforcing feasible safety constraints
Organizations	Workload, available capacity, retained expertise	Specifying whose goals, resources, and acceptable risks are represented

310 5. Domain Translations and Their Limits

The five components can organize questions across several domains, but an analogy is useful only if it leads to an explicit state, measurable quantities, and a testable dynamical model. Table 1 summarizes candidate translations. It does not assert that the same five mechanisms have been found in every
315 domain.

5.1. *Biological and ecological systems*

Physiological feedback provides motivation for analysing how multiple processes maintain functional ranges [5]. However, identifying opposing regulatory signals is insufficient to establish epistemic pull, computational drag,
320 or action efficiency as distinct state-space fields. Nor should chronic dysfunction be classified as successful adaptive homeostasis simply because it persists.

The physiological mechanisms and criteria of function must be established independently.

Ecological dynamics further illustrate why an operating regime must not
325 be conflated with a stationary point. Oscillations may occur around an equilibrium without convergence to it; their stability depends on the model. In the classical undamped Lotka–Volterra example, closed orbits are not asymptotically attracting. Other ecological models may possess attracting cycles, multiple equilibria, or transitions to different regimes. Such behaviours call
330 for analysis of invariant sets and disturbances, rather than treating all persistence as evidence of a stable equilibrium.

Ecological resilience research motivates this broader attention to dynamics [12]. Translating its mechanisms into UVIF requires explicit mappings, and the explanatory value of the original ecological model remains the base-
335 line against which the translation should be judged.

5.2. *Artificial and organizational systems*

Artificial agents provide a more direct setting because computation, sensing, action, and resource monitoring can be instrumented. Nevertheless, the agent’s reserve balance, processing budget, and environment must be mod-
340 elled together. An update that reduces a cost surrogate may still consume additional resources, and a nominally protective rule may prevent useful adaptation. Performance should therefore be measured at the whole-system level.

Organizations offer a further analogy through the allocation of limited capacity, preservation of expertise, and long-term planning. Their objectives
345 may conflict across stakeholders, and institutional persistence is not necessarily a social benefit. Organizational decision structures involving AI already

require attention to the allocation of human and automated responsibilities [19]. A UVIF application would need to expose those choices rather than conceal them behind a single organizational equilibrium.

350 **6. Relation to Cognitive Theories and Testable Hypotheses**

6.1. Cognitive interpretation

For a cognitive implementation, the state might include a belief representation, a policy, retained knowledge, and resource estimates. Epistemic pull then concerns informative sampling; selective protection concerns retention
355 and integrity; computational drag concerns the cost of inference; action efficiency concerns task performance and timeliness; and sustainability concerns future capacity to continue those activities. These are modelling assignments whose separability must be tested.

A cognitive Variational Equilibrium uses the same stationary definition
360 as Eq. (9). It is not defined as a point where one objective cannot improve without harming another: that would introduce a different, Pareto-based notion. Continuing cognition is more naturally described through trajectories in an admissible operating set, with equilibria relevant only to particular state summaries or fixed conditions.

365 *6.2. Predictive processing and active inference*

Predictive processing emphasizes hierarchical predictions and the updating of beliefs through prediction errors [20]. Active inference provides a generative-model account of perception and action, including epistemic and pragmatic contributions to policy evaluation [1, 17, 2]. Combining these con-
370 tributions in one functional does not make them indistinguishable or imply

that their behavioural balance must be fixed. That balance depends on beliefs, preferences, policies, and the model’s treatment of the environment and internal states.

UVIF makes a different organizational choice by assigning named update
375 components to regulatory demands. This may facilitate intervention and reporting, but it is not yet a derivation of a distinct class of behaviour. In particular, a resource-induced change in exploration is not by itself evidence against active inference: an active-inference model that represents resource state may also predict it. A valid comparison must identify which variables
380 and mechanisms each fitted model includes.

Likewise, a common active-inference formulation may omit a detailed account of inference expenditure without that omission being an impossibility of the framework. The pertinent comparison is between specified models with and without explicit computational and reserve accounting. Claims
385 about entire theoretical families would exceed what the present framework establishes.

6.3. Resource-rational analysis

Resource-rational analysis explains cognitive strategies in terms of expected benefits relative to computational costs [3, 4]. Human planning stud-
390 ies provide examples of how such models can be tested [9]. This literature directly anticipates the interaction between action efficiency and computational drag.

The proposed sustainability component asks whether the resource state carried between decisions changes the preferred computation policy. A per-
395 task model that omits those dynamics may fail, but an extended resource-

rational model can represent them. Irreversible outcomes can similarly be represented through absorbing states, constraints, or preferences. UVIF’s possible contribution is a convenient decomposition and independently measurable mechanisms, not the exclusive ability to represent these considerations.

6.4. *Allostasis and cognitive architectures*

Allostasis concerns predictive physiological regulation [6]. Evidence for a large-scale brain system supporting allostasis and interoception motivates examining cognition and bodily regulation together [21]. It does not establish a one-to-one mapping between a neural network and the five proposed forces.

Such a mapping would require independent measurements and interventions. Integrated cognitive architectures describe mechanisms for memory, learning, control, and action [22, 23]. UVIF is a proposed regulatory description that could be implemented within such architectures. It does not specify a complete cognitive architecture. A concrete implementation must explain how component signals are computed, how competing updates are resolved, and how the computation used by the regulatory layer itself is charged to the resource budget.

6.5. *Comparative hypotheses*

The following hypotheses concern specified implementations rather than unique predictions of UVIF as a whole. Their value lies in enabling controlled comparisons and revealing when the additional decomposition is unnecessary.

(H1) **Resource-dependent allocation.** Manipulating available computation, time, or reserve state while preserving external task informa-

420 tion and rewards should alter information acquisition when the specified model couples allocation to those resources. Compare a resource-responsive UVIF model with both fixed-weight and resource-responsive alternatives. Measure sampling decisions, response time, task performance, and resource expenditure. A failure to improve held-out prediction over equally flexible alternatives would weaken the case for the
425 proposed decomposition.

(H2) **Effects of resource carry-over.** When expenditure reduces capacity in later episodes, a model that tracks reserves should predict different allocation from an otherwise comparable model that resets resources
430 after each episode. Match the observed task and state information across agents, and compare with a long-horizon resource-rational or constrained-control model. The prediction concerns explicit carry-over dynamics, not an assumed distinction between a sustainability force and all possible long-horizon utilities.

435 (H3) **Protection under irreversible outcomes.** Compare a hard admissibility mechanism with calibrated soft penalties under the same transition model, including irreversible states. Measure violation frequency, retained task performance, and unnecessary refusal. Hard and soft formulations may coincide over a bounded range of payoffs; the experiment
440 must report that range. No behavioural observation here establishes impossibility of representation by every utility model.

(H4) **Component ablation.** Remove one implemented component while preserving observations and evaluating both fixed-parameter and re-

tuned controls. Predefine the expected consequence: reduced learning
445 under epistemic ablation, greater damage under protection ablation, or
reserve depletion under sustainability ablation. Effects need not be
monotonic or unique, because other components may compensate. An
ablation is informative only when it tests an independently meaningful
mechanism rather than a failure built into the outcome definition.

450 Section 11 specifies common evaluation requirements. This paper does
not report tests of H1–H4 or claim that they distinguish UVIF from every
member of the comparator families. The companion repository provides an
executed notebook and saved outputs for illustrative policy-grid comparisons
[24]. Its implementation maximizes a scalar score formed from the five named
455 contributions, rather than integrating Eq. (2). Its fixed-reference H1 com-
parator also freezes the environment arguments of the component functions;
this tests state responsiveness, not an impossibility of fixed-weight scalar op-
timization. The H3 search examines a particular power-penalty family over
bounded grids, which cannot establish non-equivalence to every finite mono-
460 tone penalty. These demonstrations motivate revised comparisons but are
not presented here as validation of H1–H4 or of the vector-field model.

7. Hypothetical Design Scenarios

The following scenarios illustrate how the vocabulary could guide design
and evaluation. They are not case studies with collected data, implemented
465 controllers, or demonstrated benefits. Each requires a domain-specific model
and comparison with established practice.

7.1. *Clinical decision support*

A decision-support system may have to allocate further assessment while accounting for uncertainty, clinician workload, and the consequences of delay.
470 Epistemic pull could represent the expected decision value of an additional observation; selective protection could concern integrity of the patient record and validated model components; computational drag could track inference latency and processing cost; action efficiency could represent timely decision support; and sustainability could track review capacity across a sequence of
475 cases.

Clinical test costs and burdens should not be conflated with computation costs: they belong in the explicitly defined action and resource model. Clinical eligibility, escalation rules, and patient safety requirements would be externally specified constraints. No treatment recommendation or clinical
480 benefit follows from the force metaphor. Evaluation would require clinical oversight, relevant approvals, and evidence that the model improves appropriate outcomes over validated decision-support baselines.

7.2. *Energy management*

For a resource-limited infrastructure controller, a candidate state includes
485 reserve capacity, predicted demand, equipment condition, and available computation. The proposed decomposition separates forecasting, preservation of critical components, computation expenditure, dispatch performance, and future supply capacity. Existing optimization methods can represent these quantities; UVIF would need to show that its decomposition improves monitoring, adaptation, or diagnosis.
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A useful comparison would apply identical demand sequences, reserve models, actuator limits, and forecast information to the competing controllers. Outcomes would include unmet demand, constraint violations, computational burden, and reserve depletion over a declared horizon. A
495 favourable average cost would not compensate for an unreported violation of a hard operational requirement.

7.3. Autonomous transportation

The navigation example in Section 2 can be extended to sensor allocation and memory updates. A UVIF implementation would need an explicit
500 transition model, a way to connect policy updates to vehicle controls, and a safety mechanism that respects physical braking and sensing limitations. Equilibrium-seeking dynamics alone do not ensure recovery from an obstacle or sensor failure.

Testing should separate normal operation, known disturbances, and conditions outside the assumed model. Task completion, reserve use, intervention
505 frequency, and failures should be reported together. Comparators should include resource-aware constrained planning or control, not only a controller optimized for travel speed.

7.4. Cybersecurity

510 A monitoring system may allocate inspection effort between immediate threat detection and sustained availability. The components could represent information gathering, protection of keys and logs, inspection expenditure, response effectiveness, and future monitoring capacity. A threat model must

state attacker capabilities and which assets are protected. No architectural
515 label makes a system resistant to adversarial inputs.

Ablations could examine whether an explicit capacity model reduces over-
load under repeated alerts, while retaining detection performance. Both
missed threats and disruptions to legitimate activity matter. Tuning the pro-
tective component only on attacks used for evaluation would invalidate the
520 intended comparison.

7.5. Organizational capacity

A service organization may balance information gathering, retention of
expertise, processing effort, timely service, and future staffing capacity. Here
the central modelling issue is governance: whose workload, risk, and service
525 quality count, and who may alter the constraints? A persistent organization
can still impose unacceptable costs on its members or clients.

The framework can make these trade-offs explicit, but it cannot select le-
gitimate priorities on its own. Evaluation would require stakeholder-defined
outcomes and comparisons with existing capacity-planning methods. Insti-
530 tutional longevity is not an adequate sole measure of success.

8. Comparative Methodological Positioning

Table 2 separates mathematical methods, learning frameworks, and orga-
nizational descriptions. They operate at different levels and should not be
ranked as if each were a complete alternative to the others.

Table 2: Scope of comparison. The final column identifies what a UVIF evaluation must establish, rather than assigning unsupported superiority scores.

Approach	Relevant capability	Requirement for a UVIF comparison
Constrained and multi-objective optimization	Explicit trade-offs, constraints, multiple criteria	Show value beyond relabelling objectives or adapting weights
Safe reinforcement learning	Sequential learning with risk and safety considerations	Match observations, constraints, horizons, and training resources
Active inference	Generative-model treatment of action and information acquisition	Compare fully specified resource-aware models
Resource-rational analysis	Benefits of cognition relative to computation costs	Include reserve carry-over and equivalent planning opportunities
Allostatic regulation	Anticipatory regulation of internal conditions	Test an independently measurable five-component mapping
Cognitive architectures	Integrated mechanisms for memory, learning, and action	Specify and measure the cost of the added regulatory layer
UVIF	Proposed explicit five-component decomposition	Establish identifiability, tractability, and incremental predictive value

535 *8.1. Optimization and learning as possible implementations*

Equation (13) demonstrates that one valid UVIF instance is also an optimization system. A multi-objective interpretation, a constrained policy search, or a learning-based controller may therefore implement the proposed organization. UVIF should not be distinguished solely by the use of several
540 considerations or by adaptive weighting, both of which can be represented within established mathematical frameworks.

Reinforcement learning provides methods for learning from sequential interaction, while safe RL addresses additional risk and constraint requirements [7]. The question for UVIF is whether its explicit assignment of regulatory
545 roles improves specification, transfer, or failure diagnosis under a fair comparison. A hard constraint can be a stronger protection mechanism than a soft force; describing it as external does not make it less effective.

Data-driven models likewise use explicit training objectives, even when the learned representations are not readily interpretable. Such models could
550 supply prediction, perception, or value estimates inside a UVIF implementation. The regulatory decomposition does not by itself render their representations interpretable or improve generalization.

8.2. The intended level of novelty

The proposed contribution is a modelling discipline: name the regulatory
555 demands, assign measurable state-dependent mechanisms to them, distinguish costs from constraints, and evaluate persistence together with task performance. It remains possible that a smaller decomposition, an established architecture, or a single well-specified objective will provide an equally good or better account.

560 Demonstrating novelty requires more than broad applicability. It requires
a case in which the decomposition supports a useful measurement, interven-
tion, prediction, or design decision that is difficult to obtain from a compa-
rably explicit baseline. The current paper motivates that programme and
gives a consistent template; it does not establish that the requirement has
565 been met empirically.

9. Implications for Evaluation and Interpretation

9.1. *Persistence must be measured alongside performance*

A regulatory framework should be evaluated across time, not only at a
final task score. Useful outcomes include task quality, decision latency, re-
570 source expenditure, retention of prior competence, constraint violations, and
recovery following a specified disturbance. Reporting these quantities sepa-
rately prevents an aggregate score from hiding an operationally important
failure.

For a known stationary model, distance to a stable equilibrium may help
575 characterize recovery. In a dynamic task, that distance can be misleading: a
useful action may move the system away from a frozen equilibrium, and a
failing system may have a small net update because it is stuck. Equilibrium
proximity is therefore a model-dependent diagnostic rather than a general
intelligence metric.

580 9.2. *Accounting for the regulatory layer*

Computing information gain, estimating risk, monitoring reserves, and
adapting weights all incur costs. These must be included in the same budget

used to evaluate the agent. Otherwise an apparently resource-aware design can receive an unreported computational advantage. Model calibration and
585 online operation should also be distinguished: expensive offline fitting does not have the same implications as an expensive real-time decision rule.

The five components should support diagnosis only where the mapping from observables to mechanisms is justified. A decline in performance cannot be attributed to the sustainability component merely because it occurs late
590 in a deployment. Alternative explanations include distribution shift, memory corruption, and changing task difficulty. Controlled perturbations and held-out predictions are needed to separate them.

9.3. Persistence is not a complete account of intelligence

Stability and viability are relevant to sustained cognitive performance,
595 but they do not explain all learning, reasoning, representation, or social understanding. Some intelligent actions deliberately leave a familiar operating regime. A system can also remain stable while pursuing harmful or irrelevant objectives. UVIF should therefore complement task- and context-specific accounts of cognition, rather than replace them with an unrestricted definition
600 based on equilibrium.

10. Limitations and Open Challenges

10.1. The decomposition is posited and may not be identifiable

The five components are motivated by recurring design concerns, not derived as necessary or sufficient constituents of intelligence. Other decomposi-
605 tions may be equally useful. If a vector field is added to one component and

subtracted from another, the net dynamics can remain unchanged. Observing trajectories alone consequently need not identify the components. Independent measurements, structural restrictions, or interventions are required before a force label can be given a causal interpretation.

610 Overlapping cost measures create a related problem. Energy may appear in both computational drag and sustainability, and protection may overlap with risk constraints. A model must explain whether these terms represent distinct timescales, distinct consequences, or deliberate multiple valuation. Without that explanation, apparent advantages may reflect repeated penal-
615 ties rather than a new mechanism.

10.2. The mathematical formulation is restricted

The projected dynamics assumes a continuous state, directly selectable first-order updates, and a fixed convex feasible set. Real systems may have discrete modes, nonconvex constraints, delayed observations, and limited con-
620 trol authority. Existence of an equilibrium, stability, robustness, and tracking do not follow from the force labels. The analytical example establishes those properties only for its scalar linear field and fixed parameters.

A general variational functional has not been derived. It would be misleading to infer a common principle of stationary action from the proposed
625 notation. Extensions to stochastic systems would also need probabilistic definitions of constraint satisfaction and persistence, with explicit disturbance assumptions and time horizons.

10.3. Empirical and comparative support is absent

This manuscript reports no new experiment or validated simulation comparison. The companion repository contains illustrative simulations with stipulated objectives and restricted comparators (Section 6.5). H1–H4 describe a prospective evaluation programme; they are not demonstrated effects. Neither the clinical and engineering scenarios nor the biological analogies provide independent validation. The lack of evidence for incremental value over resource-aware alternatives is the principal obstacle to presenting UVIF as an established theory or proven architecture.

The literature discussion is selective. A more comprehensive account should examine viability theory, adaptive and robust control, constrained decision processes, and related dynamical approaches to cognition in depth. Any claim of priority or formal generalization would require that comparison and a precise derivation.

10.4. Normative and cross-domain limitations

Admissibility specifications may omit important harms or encode contested priorities. Mathematical feasibility is not a substitute for domain expertise, governance, or ethical judgment. Persistent operation is not inherently desirable, and natural systems provide no automatic standard for acceptable artificial behaviour.

Broad domain translation also risks becoming unfalsifiable if every outcome can be redescribed as a change in force balance. To limit that risk, an application must specify the component measurements, parameter rules, and predicted outcomes before evaluation. A result inconsistent with those

commitments should count against that implementation, rather than being absorbed by unconstrained redefinition of its fields.

11. A Concrete Research Agenda

655 11.1. *Start with an instrumented cognitive task*

A practical first test is a sequential information-sampling task with an explicit processing budget and a resource reserve carried between episodes. The agent chooses whether to gather more information or act; both choices have recorded costs and consequences. The environment should expose ground-
660 truth transitions for verification while controlling what each agent can observe. This setting connects the cognitive allocation question to measurable system dynamics without requiring claims across every proposed domain.

Before fitting, specify the state variables, observation and transition models, component functions, normalization constants, constraints, reserve dy-
665 namics, and weight-update rules. Separate task benefit from the definition of equilibrium. Record how parameters are chosen and which information is available during training and evaluation. This specification should make it possible to reproduce each policy decision.

11.2. *Compare matched models on held-out conditions*

670 Include a fixed-weight model as a diagnostic baseline, but also compare against adaptive constrained optimization or control, resource-rational allocation with carry-over, and a resource-aware active-inference model where feasible. Give the models the same observations, action set, reserve dynamics, and operational constraints. Report parameter counts, tuning budgets,
675 training cost, and online cost.

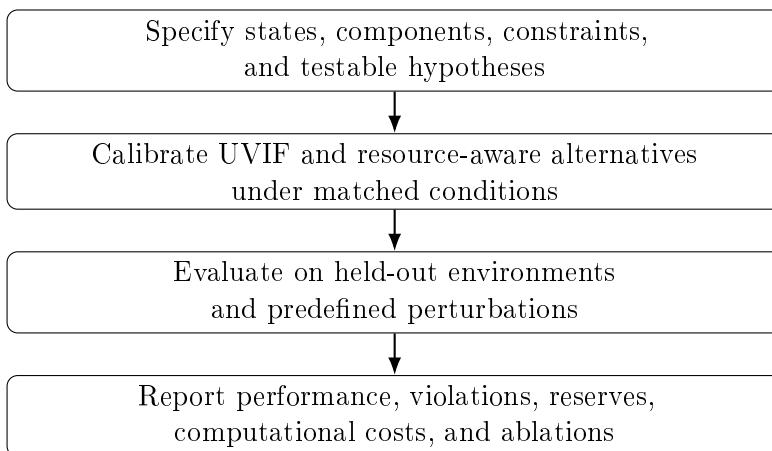


Figure 2: Proposed evaluation workflow. This is a research protocol, not a summary of completed experiments. Schematic prepared as editable TikZ code with OpenAI Codex (GPT-6) assistance.

Use disjoint calibration and evaluation conditions, varying resource scarcity, observation quality, deployment duration, and disturbance severity. Fix model selection rules before inspecting evaluation outcomes. For stochastic experiments, report independent runs, uncertainty intervals, effect sizes, and the procedure used to choose the number of runs. Human studies
 680 require a separate design with appropriate ethical review and a justified sample-size plan.

11.3. Define meaningful outcomes and negative evidence

Primary outcomes should include retained task performance, resource depletion, violation frequency, and recovery after controlled perturbations. Reserve thresholds and unacceptable events must be determined independently
 685 of the UVIF score. Measure component computation and monitoring overhead. Report all predefined outcomes, including cases in which a baseline performs better.

690 For H1–H4, specify the predicted direction and the circumstances under which no effect is expected. Compare both frozen and retuned ablations so that loss of calibration is not mistaken for evidence of a necessary component. A model whose additional fields do not improve held-out prediction, operational trade-offs, or diagnostic usefulness over a simpler matched model
695 would not justify its extra complexity in that setting.

11.4. Establish stability and reproducibility before expansion

A subsequent theoretical programme should address invariant sets, control feasibility, disturbance bounds, and tracking in the chosen implementation. A useful result would state assumptions under which a particular
700 controller maintains specified constraints for a specified class of disturbances. It would not imply universal safety or unrestricted persistence.

Release a versioned implementation, environment definitions, calibration and evaluation splits, parameter settings, random seeds, software dependencies, and scripts that regenerate all reported tables and plots. A permanent
705 archive identifier or immutable commit is preferable to a moving repository alone. Only after this focused evaluation should the framework be extended to additional domains or multi-agent systems, where individual persistence and collective welfare may conflict.

12. Conclusion

710 UVIF proposes an explicit decomposition of five regulatory demands relevant to persistent cognitive systems: information acquisition, selective protection, computation cost, effective action, and future operating capacity. The present formulation distinguishes these soft influences from hard admissibility

requirements and separates stationary equilibrium from invariant operation
715 and adaptation under changing conditions.

The analytical example demonstrates stability in one restricted model and also shows that its dynamics can minimize a scalar objective. The contribution is therefore a proposed organization of modelling responsibilities, not a demonstrated replacement for optimization. Its relationship to active
720 inference, resource-rational analysis, allostasis, and cognitive architectures is potentially complementary.

The decisive next step is a controlled comparison in which the components are independently specified, their computational costs are counted, and their predictions are tested against capable alternatives. Until such evidence is
725 available, UVIF should be treated as a conceptual research programme whose usefulness, component structure, and range of applicability remain open to revision.

Data availability

No empirical datasets are reported in this conceptual study. The ana-
730 lytical example is specified in full in Section 4.5. Companion policy-grid simulation code and saved outputs are publicly accessible in the versioned repository [24]. Those illustrative simulations are distinct from the revised vector-field formulation and do not establish the comparative hypotheses as validated findings.

735 **CRedit author contribution statement**

Habib Hamam: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

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Declaration of competing interest

745 The author declares that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Ethics statement

Not applicable. This conceptual study reports no research involving hu-
750 man participants or animals.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex (GPT-6) was used to assist with critical review, restructuring, language revision, mathematical consistency checks, LaTeX editing, and preparation of the schematic diagrams. The author is responsible for independently verifying the arguments, equations, references, and figures and for approving the final manuscript.

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